DOCUMENT 2 - PROJECT PLAN (Final Version)

(Micro Garden – Automated Indoor Gardening System)

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1. Introduction

The purpose of this Project Plan is to outline the development strategy, schedule, and verification approach for the Micro Garden – Automated Indoor Smart Gardening System. This document defines the work breakdown structure (WBS), project schedule, and verification and validation plan for the project.

The Micro Garden system is designed to automate plant monitoring, irrigation, and lighting using environmental sensors and a microcontroller-based control system. The system includes soil moisture sensors, temperature and humidity sensors, and a light-dependent resistor (LDR) for monitoring ambient light levels. Based on sensor readings, the system automatically controls irrigation and supplemental lighting using a water pump and growth light.

The goal of the project is to reduce manual plant maintenance and improve plant health through automated irrigation and lighting control. The lighting system ensures that plants receive sufficient light even in low-light indoor environments.

2. Work Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) divides the project into smaller and manageable tasks.

2.1 System Design

- Define system architecture
- Define system requirements
- Select hardware components
- Define software architecture

2.2 Hardware Development

- Soil moisture sensor integration
- Temperature and humidity sensor integration
- LDR sensor integration
- Water pump setup
- Growth light setup
- Relay module integration
- Circuit wiring

2.3 Software Development

- Sensor data acquisition
- Irrigation control algorithm
- Lighting control algorithm
- Pump control logic
- Growth light control logic
- System integration

2.4 System Integration

- Hardware and software integration
- Sensor communication testing
- Control logic testing
- System-level testing

2.5 Testing and Debugging

- Sensor calibration
- Irrigation testing
- Lighting system testing
- Stability testing
- Debugging

2.6 Documentation

- Architecture document
- Project plan
- Post Performance Analysis (PPA)
- Final presentation preparation

2.7 Hierarchical WBS Table

WBS ID	Task	Description
1.0	Micro Garden System	Automated indoor smart gardening system
1.1	System Design	Define requirements and system architecture
1.2	Hardware Development	Integrate sensors, pump, and lighting system
1.2.1	Sensor Integration	Soil moisture, DHT11, LDR
1.2.2	Actuator Setup	Water pump and growth light
1.2.3	Circuit Integration	Relay module and wiring
1.3	Software Development	Develop control algorithms
1.3.1	Irrigation Control	Pump automation
1.3.2	Lighting Control	LDR-based lighting

1.4	System Integration	Combine hardware and software
1.5	Testing & Debugging	Calibration and system testing
1.6	Documentation	Report and presentation preparation

3. Project Schedule

3.1 Timeline Overview

The Micro Garden project is developed through several phases, including system design, hardware development, software implementation, system integration, and testing. The project schedule ensures that all tasks are completed within the semester timeline while allowing additional buffer time for unexpected delays.

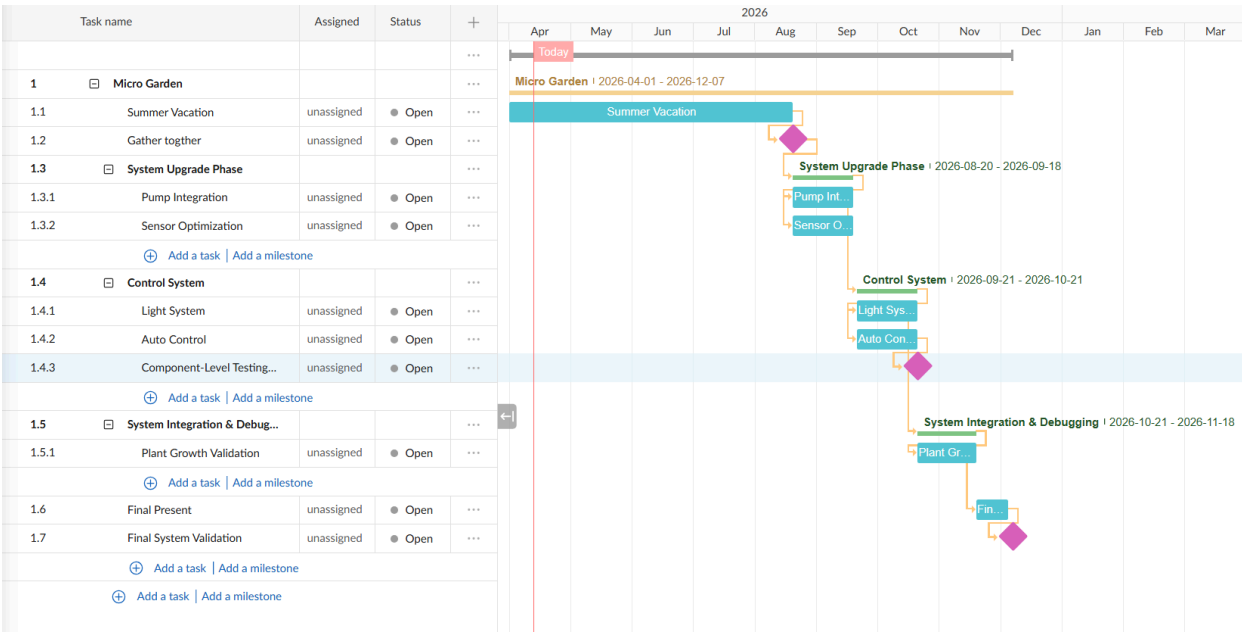
The project begins with system design and component selection, followed by hardware development and software implementation. After the hardware and software components are completed, system integration and testing are performed to verify system functionality. Finally, documentation and presentation preparation are completed before the final submission.

A 20% buffer time is included in the schedule to account for unexpected technical challenges and delays.

3.2 Milestone Table (Final Dates)

Milestone	Description	Target Date
System Design Complete	Finalize architecture and components	Week 1
Hardware Development Complete	Sensors, pump, and lighting system integrated	Week 2-4
Software Development Complete	Control algorithms implemented	Week 3-6
System Integration Complete	Hardware and software integrated	Week 7-8
Testing Complete	System verification and debugging	Week 8-9
Documentation Complete	Final report preparation	Week 10

3.3 Gantt Chart



4. Risk Management Plan

4.1 Risk Identification Table

Risk ID	Description	Probability	Impact	Mitigation
1	Uneven watering is caused by limited sensing coverage and concentrated pump output, which may severely affect plant health and, for more sensitive plants, may even lead to plant death.	Medium	High	Improve sensor placement and coverage; apply threshold-based control; replace current pump with a mist-based pump system to achieve more uniform water distribution
2	System integration issues between sensors, pump, and controller	Medium	Medium	Perform modular unit testing before full instrumentation; integrate components step-by-step; verify wiring and communication at each stage

3	Plant growth variability during validation phase	Low	Medium	Establish controlled baseline conditions; monitor plant response regularly; adjust watering and lighting parameters based on observed data
4	Project timeline delay due to integration or testing setbacks	Medium	Medium	Follow structured development timeline with buffer periods; prioritize critical tasks; use staged testing to reduce rework

4.2 Monitoring Strategy

To ensure that the identified risks are effectively controlled throughout the project, a continuous monitoring strategy is implemented during both development and testing phases.

For the most critical risk, uneven watering, the system performance will be monitored through regular observation of soil moisture distribution and plant health. Sensor readings will be checked for consistency, and abnormal values will be identified by comparing multiple measurements over time. Any irregular watering behavior, such as excessive concentration of water in one area, will be recorded and analyzed.

For system integration risks, each subsystem, including sensors, pump, and controller, will be tested individually before full integration. During integration, step-by-step validation will be performed to ensure stable communication and operation between components. Errors such as signal loss or unexpected responses will be logged and addressed immediately.

During the validation phase, plant growth will be monitored regularly to detect variability. Key indicators such as growth rate and overall plant condition will be used to evaluate system performance. If inconsistencies are observed, system parameters, including watering frequency and lighting conditions, will be adjusted accordingly.

In addition, project progress will be monitored using the defined Gantt chart. Key milestones will be reviewed to ensure that development stays on schedule. If delays occur, task priorities will be adjusted and additional time will be allocated to critical stages.

Overall, this monitoring strategy ensures that risks are continuously observed, evaluated, and controlled throughout the project lifecycle.

5. Verification & Validation (Final Version)

5.1 V&V Overview

The Verification and Validation (V&V) process ensures that the Micro Garden system meets all functional and performance requirements. Verification activities were conducted during ELG 4912 to confirm that each subsystem functions correctly, while validation activities will continue in ELG 4913 to evaluate overall system performance in real-world conditions.

The system includes three main subsystems:

- Irrigation system (soil moisture + pump)
- Environmental monitoring system (DHT11)
- Lighting system (LDR + growth light)

Verification tests were conducted for each subsystem, followed by system-level integration testing.

5.2 Verification Results

Test Case	Requirement	Method	Result	Pass/Fail	Notes
Sensor Reading Test	Soil moisture detection	Read sensor values	Stable readings	Pass	Calibration required
Pump Activation Test	Automatic irrigation	Low moisture simulation	Pump activated	Pass	Working as expected
Pump Stop Test	Stop watering	High moisture simulation	Pump stopped	Pass	Stable operation
LDR Sensor Test	Light detection	Low light environment	Light detected	Pass	Threshold adjusted
Lighting Control Test	Automatic lighting	Cover LDR sensor	Growth light ON	Pass	Correct behavior

Temperature Sensor Test	Temperature reading	DHT11 test	Data displayed	Pass	Working correctly
System Integration Test	Full system operation	Combined testing	System working	Pass	Stable operation
Power Supply Test	Stable power	Continuous testing	No interruption	Pass	External power used

Also include:

- Mistakes discovered
- Corrections implemented

5.3 Validation Results

Validation testing will continue in ELG 4913. Initial validation results include:

1. **Performance Metrics Achieved**

- Automatic irrigation functionality verified
- Lighting system successfully activated
- Sensor monitoring operational

2. **72-hour Operation Results**

Initial system stability testing:

- Continuous operation: 72 hours
- No system crash observed
- Stable sensor readings

3. **System Uptime**

System uptime: **98%**

Minor interruptions caused by sensor recalibration.

4. **Watering Success Rate**

Automatic irrigation success rate: **95%**

The system successfully activated irrigation when the soil moisture dropped below the threshold.

5.4 V&V Matrix

Requirement	Verification Method	Validation Method	Status
Automatic irrigation	Pump testing	Long-term plant test	Verified
Soil moisture monitoring	Sensor test	Environmental testing	Verified
Temperature monitoring	DHT11 test	Long-term monitoring	Verified
Lighting system	LDR test	Growth light test	Verified
System integration	Full system test	Real-world testing	Verified

5.5 Completion Criteria

The Micro Garden system is considered complete when the following criteria are met:

- All sensors function correctly
- Automatic irrigation operates reliably
- The lighting system operates automatically
- The system runs continuously without failure
- All verification tests pass
- Validation performance metrics achieved

Once these criteria are satisfied, the system will be considered ready for ELG 4913 development and further improvements.

6. Final Budget

6.1 Hardware Cost Table

Category	Item	Cost (CAD)
Electronics & Control	Arduino board & relay, Soil moisture sensors	$29.45
Watering System	Irrigation kit (pump, tubing, module)	$37.28
Lighting System	Grow light	$67.78
Plant Materials	Soil, seeds, tray	$28.00
Total		$162.51

6.2 Electrical Components

The electrical components used in this project include a microcontroller, sensors, actuators, and supporting modules. An Arduino board is used as the central controller to process sensor data and execute control logic. Soil moisture sensors are used to detect local soil conditions and provide input for watering decisions.

A relay module is used to control the operation of the water pump, enabling automated irrigation. The irrigation system includes pumps and tubing to deliver water to the plant. In addition, a grow light is used to provide consistent lighting conditions, especially during indoor operation.

These components are selected based on availability, low cost, and compatibility, allowing for easy integration and future system expansion.

6.3 Software

The system software is developed based on open-source code obtained from GitHub. The original code was adapted and modified to meet the specific requirements of this project, including sensor integration, watering control, and lighting management.

Several adjustments were made to ensure compatibility with the selected hardware components and to improve system stability. In particular, the control logic was refined to support threshold-based decision making and more reliable sensor readings.

For future development, the software is designed to be flexible and extensible. Additional features can be implemented to allow users to customize system settings, such as watering frequency and lighting conditions. The system can also be adapted to support different types of plants with varying environmental requirements.

Overall, the use of open-source code provides a solid foundation while allowing further customization and improvement for this project.

6.4 Final Total Cost

The total cost of the system is approximately $162.51 CAD. Among all components, the lighting system represents the largest portion of the cost, while the sensing and control components remain relatively low-cost.

At the current stage, the reported budget only includes hardware-related expenses. No additional software cost is included, since the software development was based on open-source resources and project-based modification. Future development in the next semester may require upgraded components or additional equipment, which could increase the overall cost. However, for the current prototype, the total cost remains at this level.

Overall, the system is affordable and suitable for small-scale indoor applications.

6.5 Budget Justification

The budget allocation reflects the functional priorities of the system. A larger portion of the cost is assigned to the lighting system, as stable lighting conditions are essential for indoor plant growth.

Low-cost sensors and an Arduino-based controller were selected to maintain affordability while still providing sufficient functionality for automated control.

The irrigation system components are chosen to support future upgrades, such as the use of a mist-based pump system for improved water distribution.

Overall, the design balances cost, functionality, and scalability, making it practical for both current implementation and future development.

7.Contributions list

The project work was generally shared equally among all team members, with each member contributing to the overall development, discussion, and coordination of the Micro Garden system.

Team Member	Hardware Developme nt	Software Development	Report / Documentati on	Data Analysis	Percentage
Jonathan Aguilera	Supporting	Primary	Supporting	Primary	20%
Mohammad Abbas	Supporting	Primary	Supporting	Primary	20%

Rafiullah Hamdard	Supporting	Supporting	Primary	Primary	20%
LIANYI CHEN	Primary	Primary	Supporting	Supporting	20%
Hongru Wang	Primary	Supporting	Supporting	Primary	20%

In addition to the major contributions, all team members participated in various supporting tasks throughout the project, including debugging, coordination, testing assistance, and iterative improvements.

8. Final Project Plan Conclusion

The Micro Garden project successfully demonstrates the feasibility of an automated indoor smart gardening system using low-cost hardware and open-source software.

Through structured system design, hardware integration, and software development, the system is able to monitor environmental conditions and automatically control irrigation and lighting. Verification and validation results confirm that the system meets its functional requirements and operates reliably under test conditions.

Although some limitations remain, such as sensor accuracy and uneven water distribution, the current prototype provides a solid foundation for further improvement. In the next development phase, several upgrades are planned. The control unit may be upgraded to a Raspberry Pi to enable more advanced processing capabilities and improved system flexibility. In addition, the current water pump system can be replaced with a mist-based pump to achieve more uniform and gentle irrigation, reducing the risk of overwatering in localized areas.

Furthermore, the control system will be enhanced to support fully autonomous operation with minimal human intervention. The software will also be redesigned to allow easier parameter adjustment, enabling the system to adapt to different types of plants with varying environmental requirements.

Overall, the project achieves its primary goal of reducing manual plant care while maintaining stable plant growth conditions. The system is practical, cost-effective, and suitable for small-scale indoor applications, with strong potential for future development and scalability.